

S.E in 3D

$$-\frac{\hbar^2}{2m} \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \left(\frac{\partial^2 \psi}{\partial \phi^2} \right) \right] + V\psi = E\psi.$$

$$\frac{1}{r} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \frac{2mr^2}{\hbar^2} [V(r) - E] = \ell(\ell+1); \quad \frac{1}{\Theta} \left[\sin \theta \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \ell(\ell+1) \sin^2 \theta \right] = m^2;$$

$$\frac{1}{Y} \left\{ \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial Y}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 Y}{\partial \phi^2} \right\} = -\ell(\ell+1). \quad \frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = -m^2.$$

$$P_\ell^m(x) \equiv (-1)^m (1-x^2)^{m/2} \left(\frac{d}{dx} \right)^m P_\ell(x), \Rightarrow \text{Associated Legendre Poly}$$

$$P_\ell(x) \equiv \frac{1}{2^\ell \ell!} \left(\frac{d}{dx} \right)^\ell (x^2-1)^\ell. \quad \begin{matrix} \text{Legendre Poly nomial} \\ P_0(x)=1, P_1(x)=x, P_2(x)=\frac{3x^2-1}{2} \end{matrix}$$

$$Y_\ell^m(\theta, \phi) = \sqrt{\frac{(2\ell+1)(\ell-m)!}{4\pi(\ell+m)!}} e^{im\phi} P_\ell^m(\cos \theta) \quad \text{Spherical Harmonics}$$

$$\int_0^{2\pi} \int_0^\pi [Y_\ell^m(\theta, \phi)]^* [Y_{\ell'}^{m'}(\theta, \phi)] \sin \theta d\theta d\phi = \delta_{\ell\ell'} \delta_{mm'}$$

$$\ell=0, 1, \dots, n-1, m=-\ell, \dots, 0, \dots, \ell$$

$$Y_{\ell}^m(\theta, \phi) = R_{\ell}(r) Y_{\ell}^m(\theta, \phi)$$

Angular Momentum

$$\vec{L} = \vec{r} \times \vec{p} = [Y p_z - Z p_y, Z p_x - X p_z, X p_y - Y p_x] \Rightarrow L_i = \epsilon_{ijk} x_j p_k$$

$$[L_i, L_j] = i\hbar \epsilon_{ijk} L_k \Rightarrow x, y, z \text{ are incompatible observables. Can't be observed simultaneously}$$

$$\text{BUT } [L^2, L_x] = [L^2, L_y] = [L^2, L_z] = 0 \Rightarrow [L^2, \vec{L}] = 0 \quad \sigma_{L_x} \sigma_{L_y} \geq \frac{\hbar}{2} |L_z|$$

$$L_{\pm} = L_x \pm iL_y \Rightarrow [L_z, L_{\pm}] = \pm \hbar L_{\pm}, [L^2, L_{\pm}] = 0, [L_{\pm}, L_z] = \pm 2L_{\pm}$$

$$\rightarrow \text{Let the state with quantum numbers } \ell, m \text{ be } |\ell, m\rangle \text{ where } \langle \theta, \phi | \ell, m \rangle = Y_\ell^m(\theta, \phi)$$

$$L^2 |\ell, m\rangle = \hbar^2 \ell(\ell+1) |\ell, m\rangle$$

$$L_z |\ell, m\rangle = m\hbar |\ell, m\rangle$$

$$L_{\pm} |\ell, m\rangle = \hbar \sqrt{\ell(\ell+1) - m(m\pm 1)} |\ell, m\pm 1\rangle$$

$$L_+ |\ell, \ell\rangle = 0, L_- |\ell, -\ell\rangle = 0 \quad \begin{matrix} \text{Raising / lowering} \\ \text{max / min level} \end{matrix}$$

$$L^2 = L_+ L_- + L_z^2 + \hbar L_z$$

$$\langle \ell, m | \hat{L} \cdot \hat{L} | \ell, m \rangle = \hbar^2 \ell(\ell+1) - m(m+1)$$

*For each ℓ there are $2\ell+1$ m values $\Rightarrow \ell=0, \frac{1}{2}, 1, \frac{3}{2}, 2, \dots$ (integer / half integers allowed for L_z formalism)

Eigenfunctions

$$\nabla = \hat{r} \frac{\partial}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{\phi} \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi}; \quad \vec{L} = (\hbar \vec{r} \times \nabla) = \mathbf{L} = -i\hbar \left(\hat{\theta} \frac{\partial}{\partial \theta} - \hat{\phi} \frac{1}{\sin \theta} \frac{\partial}{\partial \phi} \right)$$

$$\hat{\theta} = (\cos \theta \cos \phi) \hat{i} + (\cos \theta \sin \phi) \hat{j} - (\sin \theta) \hat{k}; \quad \hat{\phi} = -(\sin \phi) \hat{i} + (\cos \phi) \hat{j}.$$

$$L_x = -i\hbar \left(-\sin \phi \frac{\partial}{\partial \theta} - \cos \phi \cot \theta \frac{\partial}{\partial \phi} \right), \quad L_z = -i\hbar \frac{\partial}{\partial \phi}$$

$$L_y = -i\hbar \left(\cos \phi \frac{\partial}{\partial \theta} - \sin \phi \cot \theta \frac{\partial}{\partial \phi} \right), \quad L_{\pm} = \pm \hbar e^{\pm i\phi} \left(\frac{\partial}{\partial \theta} \pm i \cot \theta \frac{\partial}{\partial \phi} \right)$$

$$\rightarrow \text{One can recover } L^2 = -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right]$$

$$\hat{T} Y_\ell^m(\theta, \phi) = Y_\ell^m(\pi-\theta, \phi+\pi) = (-1)^m Y_\ell^m(\theta, \phi)$$

Spin

$$[S_x, S_y] = i\hbar S_z, \quad [S_y, S_z] = i\hbar S_x, \quad [S_z, S_x] = i\hbar S_y.$$

$$S^2 |s, m\rangle = \hbar^2 s(s+1) |s, m\rangle; \quad S_z |s, m\rangle = \hbar m |s, m\rangle; \quad * \text{Spin is intrinsic angular momentum}$$

$$S_{\pm} |s, m\rangle = \hbar \sqrt{s(s+1) - m(m\pm 1)} |s, (m\pm 1)\rangle$$

This time eigenvectors are not spherical harmonics $\Rightarrow s=0, \frac{1}{2}, 1, \frac{3}{2}, \dots, m_s = -s, -s+1, \dots, 0, \dots, s-1, s$

$$\begin{matrix} |s, m_s\rangle = |\frac{1}{2}, \frac{1}{2}\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ |s, m_s\rangle = |\frac{1}{2}, -\frac{1}{2}\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \end{matrix} \quad \text{Spin 1/2 particles form a complete eigenbasis for } \hat{S}^2 \text{ and } \hat{S}_z$$

$$S^2 = \frac{3\hbar^2}{4} \mathbb{I}_2, \quad S_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \frac{\hbar}{2} \sigma_z, \quad S_{\pm} = S_x \pm iS_y$$

$$S_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{\hbar}{2} \sigma_x, \quad S_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \frac{\hbar}{2} \sigma_y, \quad S_{x,y,z}^2 = \frac{\hbar^2}{4} \mathbb{I}$$

eigen spinors of $S_z \rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ with eigval. $\frac{\hbar}{2}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ with eigval. $-\frac{\hbar}{2}$

eigen spinors of $S_x \rightarrow \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$ with eigval. $\frac{\hbar}{2}, \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$ with eigval. $-\frac{\hbar}{2}$

eigen spinors of $S_y \rightarrow \begin{pmatrix} 1/\sqrt{2} \\ i/\sqrt{2} \end{pmatrix}$ with $\frac{\hbar}{2}, \begin{pmatrix} 1/\sqrt{2} \\ -i/\sqrt{2} \end{pmatrix}$ with eigval. $-\frac{\hbar}{2}$

Hydrogen Atom

$$u(r) = rR(r) \quad \frac{d}{dr} \left(r^2 \frac{du}{dr} \right) - \frac{2mr^2}{\hbar^2} [V(r) - E] R = \ell(\ell+1) R \quad \rightarrow \quad -\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + \left[V + \frac{\hbar^2}{2m} \frac{\ell(\ell+1)}{r^2} \right] u = Eu$$

For Coulomb potential $\Rightarrow \frac{d^2 u}{dr^2} = \left[1 - \frac{\rho_0}{\rho} + \frac{\ell(\ell+1)}{\rho^2} \right] u$ where $\rho = \chi r, \rho_0 = \frac{me^2}{2\pi\epsilon_0 \hbar^2 \chi} = 2a, \chi = \sqrt{-2meE}/\hbar$

$$u(\rho) = \rho^{\ell+1} e^{-\rho} Y(\rho) \Rightarrow V(\rho) = \sum_{j=0}^{\infty} c_j \rho^j, \quad c_{j+1} = \frac{2(j+\ell+1+n)}{(j+1)(j+2\ell+2)} c_j, \quad \rho = \frac{r}{a_0 n}, \quad \chi = \frac{1}{a_0 n}$$

$$E_n = -\frac{\hbar^2 \chi^2}{2m} = -\left[\frac{me}{2\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \right] \frac{1}{n^2} = -\frac{E_1}{n^2}, \quad a_0 = \frac{2\pi\epsilon_0 \hbar^2}{me^2} = \text{Bohr Radius}$$

$$Y(\rho) = \begin{pmatrix} 2\ell+1 \\ \ell! \ell! \end{pmatrix} \left(\frac{2r}{a_0 n} \right)^{\ell} \left(\frac{d}{dr} \right)^{\ell} L_{\ell-\ell}(\chi), \quad L_q(x) = (-1)^q \left(\frac{d}{dx} \right)^q L_{\ell+q}(x), \quad L_q(x) = \frac{e^x}{q!} \left(\frac{d}{dx} \right)^q [e^{-x} x^{\ell+q}]$$

$$L_0(x) = 1, \quad L_{10}(x) = 2a^{-3/2} \exp(-r/a)$$

$$L_1(x) = -x+1, \quad L_{20}(x) = \frac{1}{\sqrt{2}} a^{-3/2} \left(1 - \frac{1}{2} \frac{r}{a} \right) \exp(-r/2a)$$

$$L_2(x) = \frac{1}{2} x^2 - 2x + 1, \quad L_{21}(x) = \frac{1}{2\sqrt{6}} a^{-3/2} \left(\frac{r}{a} \right) \exp(-r/2a)$$

$$L_3(x) = -\frac{1}{6} x^3 + \frac{3}{2} x^2 - 3x + 1, \quad L_{30}(x) = \frac{2}{3\sqrt{3}} a^{-3/2} \left(1 - \frac{2}{3} \frac{r}{a} + \frac{2}{27} \left(\frac{r}{a} \right)^2 \right) \exp(-r/3a)$$

$$L_4(x) = \frac{1}{24} x^4 - \frac{3}{2} x^3 + 3x^2 - 4x + 1, \quad L_{31}(x) = \frac{8}{27\sqrt{6}} a^{-3/2} \left(1 - \frac{1}{6} \frac{r}{a} \right) \left(\frac{r}{a} \right) \exp(-r/3a)$$

$$L_5(x) = -\frac{1}{120} x^5 + \frac{5}{24} x^4 - \frac{5}{3} x^3 + 5x^2 - 5x + 1, \quad L_{32}(x) = \frac{4}{81\sqrt{30}} a^{-3/2} \left(\frac{r}{a} \right)^2 \exp(-r/3a)$$

$$L_0^0(x) = 1, \quad L_0^1(x) = 1$$

$$L_1^0(x) = -x+1, \quad L_1^1(x) = -x+3$$

$$L_2^0(x) = \frac{1}{2} x^2 - 2x + 1, \quad L_2^1(x) = \frac{1}{2} x^2 - 4x + 6$$

$$L_2^2(x) = 1, \quad L_2^3(x) = 1$$

$$L_3^0(x) = -x+2, \quad L_3^1(x) = -x+4$$

$$\Psi_{\ell m} = \left[\frac{(2\ell)!}{(nq)!} \frac{(n-\ell-1)!}{2n(n+\ell)!} \right] e^{-r/na} \left(\frac{2r}{na} \right)^{\ell} \left[L_{\ell-\ell} \left(\frac{2r}{na} \right) \right] Y_{\ell}^m(\theta, \phi)$$

Electron Spin: $\vec{H} = -\vec{\mu} \cdot \vec{B}, \vec{\mu} = g \frac{q}{2m_e} \vec{S} = \frac{e}{m_e} \vec{S}, \vec{F} = \vec{\mu} \cdot \frac{\partial \mathcal{H}}{\partial \vec{S}} \Rightarrow \text{Stern-Gerlach}$

Full electron description: $\Psi(\vec{r}) = \chi \Rightarrow$ Full Hilbert sp. spanned by $\{ \ell, m, \frac{1}{2} \}$ vectors
For 2 non-interacting particles $\Rightarrow 40$ space with basis $|m_{s1}\rangle |m_{s2}\rangle = |\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle$
 $\hookrightarrow$ Total angular momentum $S = \vec{S}_1 \otimes \mathbb{I} + \mathbb{I} \otimes \vec{S}_2 \Rightarrow S_z |\uparrow\uparrow\rangle = 2\frac{\hbar}{2} |\uparrow\uparrow\rangle \Rightarrow \text{eval } \hbar$
 $S_z |\downarrow\downarrow\rangle = -2\frac{\hbar}{2} |\downarrow\downarrow\rangle \Rightarrow \text{eval } -\hbar$
 $S_z |\uparrow\downarrow\rangle = (\frac{\hbar}{2} - \frac{\hbar}{2}) |\uparrow\downarrow\rangle \Rightarrow \text{eval } 0$

Triplet State $S=1, M_s = -1, 0, 1$

$$\begin{matrix} |\uparrow, \uparrow\rangle = |\uparrow\uparrow\rangle \\ |\uparrow, \downarrow\rangle = |\downarrow\downarrow\rangle \\ |1, 0\rangle = \frac{1}{\sqrt{2}} [|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle] \end{matrix} \quad \left. \begin{matrix} P_{12} |\Psi_{\text{trip}}\rangle = |\Psi_{\text{trip}}\rangle \\ \text{Symmetric} \end{matrix} \right\} \quad \left. \begin{matrix} |0, 0\rangle = \frac{1}{\sqrt{2}} [|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle] \\ \text{antisym} \end{matrix} \right\} \quad \left. \begin{matrix} \text{Singlet State } (S=0) \\ |0, 0\rangle = \frac{1}{\sqrt{2}} [|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle] \\ \text{antisymmetric} \end{matrix} \right\}$$

Bell States

$$\begin{matrix} |\Phi_0\rangle = \frac{1}{\sqrt{2}} [|\uparrow\uparrow\rangle + |\downarrow\downarrow\rangle] \\ |\Phi_1\rangle = \frac{1}{\sqrt{2}} [|\uparrow\uparrow\rangle + |\downarrow\downarrow\rangle] \\ |\Phi_2\rangle = \frac{1}{\sqrt{2}} [|\uparrow\uparrow\rangle - |\downarrow\downarrow\rangle] \\ |\Phi_3\rangle = \frac{1}{\sqrt{2}} [|\uparrow\uparrow\rangle - |\downarrow\downarrow\rangle] \end{matrix} \quad \left. \begin{matrix} |+\rangle = \frac{1}{\sqrt{2}} [|\Phi_0\rangle + |\Phi_3\rangle] \\ |-\rangle = \frac{1}{\sqrt{2}} [|\Phi_0\rangle - |\Phi_3\rangle] \\ |+\rangle = \frac{1}{\sqrt{2}} [|\Phi_1\rangle + |\Phi_2\rangle] \\ |-\rangle = \frac{1}{\sqrt{2}} [|\Phi_1\rangle - |\Phi_2\rangle] \end{matrix} \right\} \quad \left. \begin{matrix} |\Phi_0\rangle = (1 \otimes \sigma_z) |\Phi_0\rangle \\ |\Phi_1\rangle = (1 \otimes \sigma_z) |\Phi_1\rangle \\ |\Phi_2\rangle = (1 \otimes \sigma_z) |\Phi_2\rangle \\ |\Phi_3\rangle = (1 \otimes \sigma_z) |\Phi_3\rangle \end{matrix} \right\}$$

Source & entangled Bell $\rightarrow$ Alice Measure AC $\begin{pmatrix} 0 & 1 & 2 & 3 \end{pmatrix}$ $\rightarrow$ Bob $\sigma_z \sigma_z \sigma_z \sigma_z$

Suppose we have 3-spin th particles $|\Phi_0\rangle_{AB}$ and $|\Psi\rangle_C = \alpha|\uparrow\rangle_C + \beta|\downarrow\rangle_C$
 $|\Psi\rangle_{ABC} = |\Phi_0\rangle_{AB} \otimes |\Psi\rangle_C = \sum |\Phi_m\rangle_{AC} \otimes |\Psi\rangle_B$ where $|\Psi_B\rangle = \alpha|\uparrow\rangle_B + \beta|\downarrow\rangle_B$
 $\hookrightarrow$ If Alice measure $|\Phi_2\rangle_{AC}$ for example, she tells Bob to do $\sigma_z \Rightarrow \sigma_z(\sigma_z |\Psi_B\rangle) = |\Psi_B\rangle$

Distinguishable particles $\Rightarrow$ 2 different particles: $\Psi_0 = \Psi_A(x_1) \Psi_B(x_2)$ $\rightarrow$ we can have $a=b$
Bosons $\Rightarrow$ identical, symmetric, integer spin (photon, He): $\Psi_B = \frac{1}{\sqrt{2}} [\Psi_A(x_1) \Psi_B(x_2) + \Psi_B(x_1) \Psi_A(x_2)]$
Fermions $\Rightarrow$ identical, antisymm, $\frac{1}{2}$ integer spin ($e, p, {}^3\text{He}$): $\Psi_F = \frac{1}{\sqrt{2}} [\Psi_A(x_1) \Psi_B(x_2) - \Psi_B(x_1) \Psi_A(x_2)]$

Exchange operator: $X_{12} \Psi(q_1, q_2) = \Psi(q_2, q_1) \Rightarrow [X_{12}, H] = 0$ for identical particles $\Rightarrow X_{12}^\dagger = X_{12}$
If $\Psi_N(q_1, \dots, q_N)$ is indistinguishable $X_{ij} \Psi_N(q_1, \dots, q_N) = \pm \Psi_N(q_1, \dots, q_N)$ for any i, j
 $\langle x_1 - x_2 \rangle_0^2 = \langle x_1^2 \rangle + \langle x_2^2 \rangle - 2 \langle x_1 x_2 \rangle = 2 \langle x_1^2 \rangle + 2 \langle x_2^2 \rangle - 2 \langle x_1 x_2 \rangle = \langle (\Delta x)^2 \rangle_D$
 $\langle (x_1 - x_2)^2 \rangle_{\pm} = \langle (\Delta x)^2 \rangle_D \mp 2 | \langle x_1 x_2 \rangle |^2$ where $\langle x \rangle_{ab} = \int x \Psi_a^*(x) \Psi_b(x) dx$

Hoxby-Mandel: $U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ $\rightarrow$ exchange force: measure of indist. / spatial overlap
 $U \frac{1}{\sqrt{2}} [|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle] = \frac{1}{\sqrt{2}} [|\uparrow\uparrow\rangle - |\downarrow\downarrow\rangle] \Rightarrow$ bunching

Harmonic Oscillator

$$V(x) = \frac{1}{2} kx^2 = \frac{1}{2} m\omega^2 x^2$$

Let $\psi = \frac{x}{x_0} = \sqrt{\frac{m\omega}{\hbar}} x \Rightarrow$ TISE: $-\frac{\hbar^2}{2m} \psi'' + \frac{1}{2} m\omega^2 x^2 \psi = E\psi \Rightarrow \frac{d^2\psi}{dy^2} + (\epsilon - y^2)\psi = 0$
 $E_n = \hbar\omega(n + \frac{1}{2})$, $\psi_n(y) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n(y) e^{-y^2/2}$, $\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n\left(\sqrt{\frac{m\omega}{\hbar}} x\right) e^{-\frac{m\omega}{2\hbar} x^2}$
 $\frac{dH_n}{dy} = 2nH_{n-1}(y)$, $H_n(y) = (-1)^n e^{y^2} \frac{d^n}{dy^n} e^{-y^2}$, $H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x)$
 $\int dx H_n(x) H_m(x) e^{-x^2} = \sqrt{\pi} 2^n n! \delta_{nm}$ | $H_0(x)=1$, $H_1(x)=2x$, $H_2(x)=4x^2-2$, $H_3(x)=8x^3-12x$
 $\langle x \rangle = \langle p \rangle = 0$, $\langle x^2 \rangle = \frac{\hbar}{2m\omega} (2n+1)$, $\langle p^2 \rangle = \frac{\hbar m\omega}{2} (2n+1)$, $\langle V \rangle = \langle T \rangle = \frac{\hbar\omega}{2} (n + \frac{1}{2})$

Raising/Lowering Operators

$X = \sqrt{\frac{\hbar}{m\omega}} \hat{x}$, $P = \sqrt{\frac{\hbar}{m\omega}} \hat{p}$, $H = \frac{1}{2} (P^2 + X^2)$, $[\hat{x}, \hat{p}] = i\hbar$, $[X, P] = i$
 $a = \frac{1}{\sqrt{2}} (X + iP)$ not observable since $a^\dagger$ is NOT Hermitian
 $a^\dagger = \frac{1}{\sqrt{2}} (X - iP)$ NOT Hermitian
 $\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (a + a^\dagger)$, $a^\dagger a = n+1 = \frac{H}{\hbar\omega} + \frac{1}{2}$
 $\hat{p} = -i\sqrt{\frac{\hbar m\omega}{2}} (a - a^\dagger)$, $a^\dagger a = n = \frac{H}{\hbar\omega} - \frac{1}{2}$ } $[a, a^\dagger] = 1$

$a|n\rangle = \sqrt{n}|n-1\rangle$, $\hat{n}|n\rangle = n|n\rangle$, $[a, \hat{n}] = a$, $a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$, $\hat{n}|n\rangle = n|n\rangle$, $[a^\dagger, \hat{n}] = -a^\dagger$
 $a|0\rangle = 0$, $a^\dagger|0\rangle = \sqrt{1}|1\rangle$, $|0\rangle = \frac{1}{\sqrt{n!}} a^n |1\rangle$

ISW

$V(x) = \begin{cases} 0, & 0 \leq x < a \\ \infty, & \text{o.w.} \end{cases}$, $\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$, $E_n = \frac{\hbar^2 \pi^2 n^2}{2ma^2}$
 $\hookrightarrow$ Even/odd w.r.t. $x=a/2$ for even/odd n
 $\hookrightarrow$ Each state ψ_n has $n-1$ nodes
 $\hookrightarrow$ Complete orthonormal set $\int_0^a \psi_n(x) \psi_m(x) dx = \delta_{nm}$

Ehrenfest: $\frac{d\langle x \rangle}{dt} = \frac{\langle p \rangle}{m}$, $\frac{d\langle p \rangle}{dt} = -\langle \frac{\partial V}{\partial x} \rangle$

Generalized Ehrenfest:

$\frac{d\langle Q \rangle}{dt} = \frac{i}{\hbar} \langle [H, Q] \rangle + \langle \frac{\partial Q}{\partial t} \rangle$

Position Eigenfunctions

$g_{x_0}(x) = \delta(x - x_0)$, $x g_{x_0}(x) = x_0 g_{x_0}(x) \Rightarrow \psi(x) = \int dx x_0 \delta(x - x_0) \psi(x_0)$

$\hookrightarrow$ continuous orthonorm. for position eigenstates: $\int \delta(x-a) \delta(x-b) dx = \delta(a-b)$

* $\delta(x-x_0), e^{-ikx}$ are duals of each other

Operators

$\langle Q^\dagger f | g \rangle = \langle f | Q g \rangle \Rightarrow Q^\dagger$ adjoint of Q , If $Q^\dagger = Q \Rightarrow$ Hermitian

* Orthonormal eigfunc. of Hermitian Q form an orthonormal basis
 * $\langle f | g \rangle^* = \langle g | f \rangle$ $\hookrightarrow$ real eigvals $\hookrightarrow$ orthogonal eigenvectors $\hookrightarrow \langle f | Q | f \rangle \in \mathbb{R}$ for them

$\rightarrow$ Measurement postulate: $\psi = \sum c_n | \psi_n \rangle$ where $| \psi_n \rangle$ is energy eigenbasis
 $\hookrightarrow$ We measure E_n (eigenval of $\hat{H}$) with prob. $|c_n|^2$

$Q | \psi \rangle = \sum c_n Q | \psi_n \rangle$, $\langle \psi | Q | \psi \rangle = \sum c_n \langle \psi_n | Q | \psi_n \rangle = \sum c_n E_n = \langle E \rangle = Q_{cl}$

$\left(\frac{\partial}{\partial x}\right)^\dagger = -\frac{\partial}{\partial x} \Rightarrow \left(i\hbar \frac{\partial}{\partial x}\right)^\dagger = -i\hbar \frac{\partial}{\partial x}$, all observables are Hermitian

Commutator of 2 Hermitian ops. is Anti-Hermitian

If A, B compatible $\Rightarrow$ they share the same eigenvectors

Generalized Uncertainty Principle: $\sigma_A \sigma_B \geq \frac{1}{2} | \langle [A, B] \rangle |$

$\int_{-\infty}^{\infty} dx e^{-ax^2} = I_0(a) = \sqrt{\frac{\pi}{a}}$, $\int_{-\infty}^{\infty} dx x e^{-ax^2} = I_1(a) = \frac{1}{2a}$

$I_2(a) = \frac{1}{2a} \sqrt{\frac{\pi}{a}}$, $I_3(a) = \frac{1}{2a^2}$, $I_4(a) = \frac{3}{4a^3}$, $I_5(a) = \frac{15}{8a^4}$

$x = r \sin\theta \cos\phi$, $\hat{\phi} = \hat{y} \cos\phi - \hat{x} \sin\phi$
 $y = r \sin\theta \sin\phi$, $\hat{\theta} = \hat{x} \cos\phi \cos\theta + \hat{y} \sin\phi \cos\theta - \hat{z} \sin\theta$
 $z = r \cos\theta$, $\hat{r} = \hat{z} \cos\theta + \hat{x} \sin\theta \cos\phi + \hat{y} \sin\theta \sin\phi$

$L_x = i\hbar \left(\sin\phi \frac{\partial}{\partial \theta} + \cot\theta \cos\phi \frac{\partial}{\partial \phi} \right)$, $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$, $\sum_{n=1,3,5,\dots}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{8}$

$L_x = i\hbar \left(-\cos\phi \frac{\partial}{\partial \theta} + \cot\theta \sin\phi \frac{\partial}{\partial \phi} \right)$
 $L_z = -i\hbar \frac{\partial}{\partial \phi}$
 $\int \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi x}{a}\right) dx = \frac{a}{2} \delta_{nm}$
 $\int \frac{\sin x}{x} dx = \pi/2$, $\int \cos x e^{-ax^2} dx = \sqrt{\frac{\pi}{a}} e^{-b^2/4a}$

Scattering

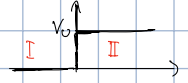
$E < V(\pm\infty) \Rightarrow$ bound state, $E > V(\pm\infty) \Rightarrow$ scattering state

Prob. Current: $\frac{d}{dt} \int dx |\psi(x,t)|^2 \Rightarrow \int dx \frac{\partial}{\partial t} |\psi(x,t)|^2 = \frac{i\hbar}{2m} \int dx \frac{\partial}{\partial x} \left[\psi^* \frac{\partial \psi}{\partial x} - \psi \frac{\partial \psi^*}{\partial x} \right] = 0$
 $\frac{\partial \rho}{\partial t} + \frac{\partial j}{\partial x} = 0$, $j = \frac{i\hbar}{2m} \left[\psi^* \nabla \psi - \psi \nabla \psi^* \right] = \frac{\hbar}{m} \text{Im} \left(\psi^* \frac{\partial \psi}{\partial x} \right)$

Step Potential

Case I: $E > V_0$

$\psi(x) = \begin{cases} A e^{ikx} + B e^{-ikx}, & x < 0 \\ C e^{ikx}, & x > 0 \end{cases}$, $k = \sqrt{\frac{2mE}{\hbar^2}}$, $\tilde{k} = \sqrt{\frac{2m(E-V_0)}{\hbar^2}}$



$J_A = \frac{\hbar k}{m} |A|^2$, $J_B = -\frac{\hbar k}{m} |B|^2$, $J_C = \frac{\hbar \tilde{k}}{m} |C|^2 \Rightarrow J(x) = \begin{cases} J_C, & x > 0 \\ J_A + J_B, & x < 0 \end{cases}$
 $\frac{\hbar k}{m} (|A|^2 - |B|^2) = \frac{\hbar \tilde{k}}{m} |C|^2 \Rightarrow \frac{|A|^2 - |B|^2}{|A|^2} = \frac{\tilde{k}}{k}$ as expected

$R = (J_B/J_A) = |B|^2/|A|^2 = \left(\frac{k-\tilde{k}}{k+\tilde{k}}\right)^2$, $T = \frac{\tilde{k}}{k} \frac{|C|^2}{|A|^2} = \frac{4k\tilde{k}}{(k+\tilde{k})^2} \Rightarrow T+R=1$

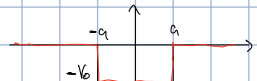
Case II: $E < V_0$

$\psi(x) = \begin{cases} A e^{ikx} + B e^{-ikx}, & x < 0 \\ C e^{-\kappa x}, & x > 0 \end{cases}$, $B = \frac{k-i\kappa}{k+i\kappa} A$, $C = \frac{2\kappa}{k+i\kappa} A$
 phase-shift upon refl'd wave $\frac{k-i\kappa}{k+i\kappa} = e^{-2i\delta(E)}$, $\delta(E) = \tan^{-1} \left[\frac{V_0}{E} \right]^{1/2}$
 $\kappa = \sqrt{2m(V_0-E)}/\hbar$

* $J_C = 0$, $J_B = J_A \Rightarrow R=1, T=0$

Finite Square Well

$V_0(x) = \begin{cases} -V_0, & |x| < a \\ 0, & \text{o.w.} \end{cases}$



Case I: $(E < 0)$ Bound States

$\psi(x) = \begin{cases} F e^{-\kappa x}, & x > a \\ D \cos(kx), & 0 < x < a \\ \psi(-x), & x < 0 \end{cases}$, $\tan(z) = \sqrt{\left(\frac{z_0}{z}\right)^2 - 1}$, $z = ka$, $z_0 = \sqrt{2mV_0} (a/\hbar)$
 $\kappa = \sqrt{2mE}/\hbar$, $k = \sqrt{2m(E+V_0)}/\hbar \Rightarrow |z_0|^2 = z^2 + \kappa^2 a^2$

$\hookrightarrow$ Finite # of bound eigenstates. $\Rightarrow$ not complete eigenbasis

Case II ($E > 0$): Scattering States

$k = \sqrt{2m(E+V_0)}/\hbar$, $\tilde{k} = \sqrt{2mE}/\hbar$

$\psi(x) = \begin{cases} A e^{ikx} + B e^{-ikx}, & x < -a \\ C e^{ikx} + D e^{-ikx}, & -a < x < a \\ F e^{ikx}, & x > a \end{cases}$, $B = -\frac{\sin(2ka)}{2k\tilde{k}} (k^2 - \tilde{k}^2) F$
 $F = \frac{e^{-2ika} A}{\cos(2ka) - i \left(\frac{k^2 + \tilde{k}^2}{2k\tilde{k}} \right) \sin(2ka)}$

$T = |F|^2/|A|^2 = \left[1 + \frac{V_0}{4E(E+V_0)} \sin^2 \left(\frac{2a}{\hbar} \sqrt{2m(E+V_0)} \right) \right]^{-1}$
 $E, E_n = \frac{\hbar^2 \pi^2 k^2}{2m(2a)^2} - V_0 \Rightarrow$ energies at which $T=1, R=0$

Delta Potential Well $\Rightarrow V(x) = -\alpha \delta(x)$

Case I ($E < 0$): Bound States

Using S.E. $\Delta \left(\frac{\psi}{\partial x} \right) = \frac{\partial \psi}{\partial x} \Big|_0^+ - \frac{\partial \psi}{\partial x} \Big|_0^- = -\frac{2m\alpha}{\hbar^2} \psi(0)$
 $\psi(x) = \begin{cases} B e^{-\kappa x}, & x > 0 \\ B e^{\kappa x}, & x < 0 \end{cases}$, $B = \sqrt{\frac{m\alpha}{\hbar^2}}$ from normalization
 $E = -\frac{\hbar^2 \kappa^2}{2m} = -\frac{m\alpha^2}{2\hbar^2} \Rightarrow \psi(x) = \sqrt{\frac{m\alpha}{\hbar^2}} e^{-m\alpha|x|/\hbar^2}$

Case II ($E > 0$): Scattering States

$\psi(x) = \begin{cases} A e^{ikx} + B e^{-ikx}, & x < 0 \\ F e^{ikx}, & x > 0 \end{cases}$, $B = \frac{i\beta}{1-i\beta} A$, $F = \frac{1}{1-i\beta} A$, $\beta = \frac{m\alpha}{\hbar^2 k}$
 $T = \frac{1}{1+\beta^2} = \frac{1}{1+(m\alpha/2\hbar^2 E)}$, $R = \frac{\beta^2}{1+\beta^2} = \frac{\beta^2}{1+(m\alpha/2\hbar^2 E)}$

Stationary Wave Approx. $\Rightarrow$ when oscillatory part changes rapidly

$\frac{d}{dk} [kx - \omega(k)t] = 0 \Rightarrow$ S.P.A.: $x - \frac{d\omega}{dk} t = 0 \Rightarrow$ max. constructive interference
 We demand k_0 to be peak of our momentum wavefunc $\Rightarrow x = \frac{\hbar k_0}{m} t = V(k_0)t$

Free Particle

$H = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \Rightarrow \psi_k(x) = C e^{ikx} + D e^{-ikx}$, $E_k = \frac{\hbar^2 k^2}{2m} \Rightarrow \psi_k(x) = C e^{ikx}$, ψ_{-k} are degenerate

ψ are not normalizable!

$f(x) = \int_{-\infty}^{\infty} \frac{dk}{\sqrt{2\pi}} F(k) e^{ikx} \Leftrightarrow F(k) = \int_{-\infty}^{\infty} \frac{dx}{\sqrt{2\pi}} f(x) e^{-ikx}$
 Defining $\phi_k(x) = \frac{e^{ikx}}{\sqrt{2\pi}}$ with $\langle \phi_k, \phi_{k'} \rangle = \delta(k-k') \Rightarrow$

$\psi(x) = \int_{-\infty}^{\infty} dk \phi_k(x) \tilde{\psi}(k)$
 $\tilde{\psi}(k) = \int_{-\infty}^{\infty} dx \phi_k^*(x) \psi(x)$

$\hookrightarrow$ One can normalize $\int dx |\psi(x)|^2 = \int dk |\tilde{\psi}(k)|^2 = 1$

$\hookrightarrow$ Prob. of measuring momentum b/w k and $k+dk: |\tilde{\psi}(k)|^2 dk$

